Population Genetics Summary 1: Allele and Genotypic Frequencies & Hardy-Weinberg Equilibrium

1. Allele Frequency

Definition: Allele frequency is the proportion of a specific allele (variant of a gene) among all copies of that gene in a population. It represents how common an allele is in the gene pool.

- Range: 0 to 1 (or 0% to 100%).
- Sum: The frequencies of all alleles for a given gene in a population will always add up to 1.

2. Genotypic Frequency

Definition: Genotypic frequency is the proportion of a specific genotype (the combination of two alleles) in a population. It represents how common a particular genetic combination is among individuals.

- Range: 0 to 1.
- Sum: The frequencies of all possible genotypes for a given gene in a population will always add up to 1.

3. Example Calculation from Genotype Numbers

Let's consider a population of 1000 pea plants. A single gene controls flower colour with two alleles:
- A (dominant) = Purple flowers
- a (recessive) = White flowers

In our population, we count the following genotypes:
- AA (Purple): 360 individuals
- Aa (Purple): 480 individuals
- aa (White): 160 individuals

Total individuals (N) = 360 + 480 + 160 = 1000

Step 1: Calculate Genotypic Frequencies

Genotypic frequency = (Number of individuals with a specific genotype) / (Total number of individuals)

- Frequency of AA = 360 / 1000 = 0.36
- Frequency of Aa = 480 / 1000 = 0.48
- Frequency of aa = 160 / 1000 = 0.16

Check: 0.36 + 0.48 + 0.16 = 1.0

Step 2: Calculate Allele Frequencies

To find the frequency of an allele, we count how many times it appears in the population. Each diploid individual carries two alleles for the gene.
- Total number of alleles in the population = 2 × (Total individuals) = 2 × 1000 = 2000 alleles.

Method 1: Counting Alleles
- Frequency of A (p):
    - Every AA individual contributes 2 A alleles.
    - Every Aa individual contributes 1 A allele.
    - Number of A alleles = (2 × number of AA) + (1 × number of Aa)
    - Number of A alleles = (2 × 360) + (1 × 480) = 720 + 480 = 1200
    - Frequency of A (p) = (Number of A alleles) / (Total alleles) = 1200 / 2000 = 0.6

- Frequency of a (q):
    - Every Aa individual contributes 1 a allele.
    - Every aa individual contributes 2 a alleles.
    - Number of a alleles = (1 × number of Aa) + (2 × number of aa)
    - Number of a alleles = (1 × 480) + (2 × 160) = 480 + 320 = 800
    - Frequency of a (q) = (Number of a alleles) / (Total alleles) = 800 / 2000 = 0.4

OR Simply, frequency of a = (q) = (1-p)

Method 2: Using Genotypic Frequencies (Shortcut)
- p = freq(AA) + (1/2) × freq(Aa)
    - p = 0.36 + (0.5 × 0.48) = 0.36 + 0.24 = 0.6
- q = freq(aa) + (1/2) × freq(Aa)
    - q = 0.16 + (0.5 × 0.48) = 0.16 + 0.24 = 0.4

Check: p + q = 0.6 + 0.4 = 1.0

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4. Hardy-Weinberg Equilibrium (HWE)

Statement: The Hardy-Weinberg Equilibrium is a fundamental principle in population genetics that states that in a large, random mating population with no selection, no mutation, and no migration, allele and genotypic frequencies will remain constant from generation to generation.

This principle describes a theoretical baseline against which we can measure evolution. If the frequencies change, it indicates that evolution is occurring.

HWE is always stated with respect to a certain locus. It is possible that within the same population, one locus is at HWE while another locus is evolving.

Explicit Assumptions of HWE:
For a population to be in Hardy-Weinberg Equilibrium at a particular locus, all of the following conditions must be met with respect to the locus in question:
1. No Mutation.
2. No Selection: All genotypes have an equal chance of survival and reproduction (no fitness differences).
3. Large Population Size: The model assumes the population to be infinitely large.
4. No Gene Flow (Migration): No individuals enter or leave the population, so no alleles are added or removed.
5. Random Mating: Individuals pair by chance, not according to their genotype or phenotype for the trait in question.

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5. Properties of a Population at HWE

When all assumptions are met and a population is in Hardy-Weinberg Equilibrium:

(a) Relationship between Allele and Genotype Frequency:
The genotypic frequencies are a simple binomial expansion of the allele frequencies.
- If `p` = frequency of allele A
- And `q` = frequency of allele a (where $p + q = 1$)
- Then, after one generation of random mating, the genotypic frequencies will be:
  - p^2 = frequency of AA genotype
  - $2pq$ = frequency of Aa genotype
  - q^2 = frequency of aa genotype
  - (This is the Hardy-Weinberg equation: $p^2 + 2pq + q^2 = 1$)

(b) Allele Frequency in Males and Females:
The allele frequencies are the same in both males and females.
For Autosomal Loci: Even if at the start, the allele frequencies in males and females are different, they become equal after one generation of random mating.
For X-linked loci: If at the start, the allele frequencies in males and females are different, Allele frequencies will take a long time to become equal in males and females.

(c) Genotype Frequency in Males and Females:
For Autosomal Loci: The genotype frequencies become the same in both males and females after one generation of random mating and remain un-changed thereafter.
For X-linked loci: Genotype frequencies in males and females will not be the same.

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6. Numerical Examples Using HWE

Example 1: Calculating Carrier Frequency (Autosomal Recessive Disease)

Cystic fibrosis (CF) is an autosomal recessive disorder caused by a mutation in the CFTR gene. In a certain population at HWE at this locus, the frequency of affected individuals (genotype `aa`) is 1 in 2500. Find the frequency of carriers.

- Step 1: Find q (frequency of the disease allele).
  1 in 2500 ($q^2 = 0.0004$)
  - $q = \sqrt{(q^2)} = \sqrt{0.0004} = 0.02$
- Step 2: Find p (frequency of the normal allele).
  - $p = 1 - q = 1 - 0.02 = 0.98$
- Step 3: Find 2pq (frequency of carriers, genotype `Aa`).
  - $2pq = 2 \times (0.98) \times (0.02) = 0.0392$

Interpretation: Approximately 3.92% of the population (or about 1 in 25 people) are carriers for cystic fibrosis, meaning they have one copy of the allele but do not have the disease.

Example 2: Determining if a Population is in HWE

We return to our original population of 1000 pea plants. We already calculated the observed genotypic frequencies:
- AA = 0.36
- Aa = 0.48
- aa = 0.16
- Allele frequencies: p = 0.6, q = 0.4

Is the population at HWE at this locus?

Step 1: Calculate the expected genotypic frequencies under HWE.
- Expected AA = $p^2 = (0.6)^2 = 0.36$
- Expected Aa = $2pq = 2 \times (0.6) \times (0.4) = 0.48$
- Expected aa = $q^2 = (0.4)^2 = 0.16$

Step 2: Compare observed vs. expected.
The expected genotypic frequencies (0.36, 0.48, 0.16) match our observed frequencies perfectly.

Conclusion: This population appears to be in Hardy-Weinberg Equilibrium for this gene. No evolution is occurring at this locus.

Example 3: A Population Not in HWE

Imagine a different population of 1000 pea plants with the following observed counts:
- AA = 500, Aa = 0, aa = 500
- Genotypic frequencies: AA=0.5, Aa=0, aa=0.5

Step 1: Calculate allele frequencies.
- p = freq(AA) + 1/2 freq(Aa) = 0.5 + 0 = 0.5
- q = freq(aa) + 1/2 freq(Aa) = 0.5 + 0 = 0.5

Step 2: Calculate expected HWE frequencies.
- Expected AA = $p^2 = (0.5)^2 = 0.25$
- Expected Aa = $2pq = 2 \times 0.5 \times 0.5 = 0.50$
- Expected aa = $q^2 = (0.5)^2 = 0.25$

Step 3: Compare observed vs. expected.
- Observed: AA=0.5, Aa=0.0, aa=0.5
- Expected: AA=0.25, Aa=0.50, aa=0.25

The observed frequencies are very different from the expected ones (we see a complete lack of heterozygotes).

Conclusion: This population is not in Hardy-Weinberg Equilibrium. The assumptions must be violated. A likely cause here could be many including non-random mating (e.g., assortative mating, where purple flowers only mate with purple flowers, and white with white) or strong selection against heterozygotes.